Dylan Joseph

Computer Vision & Full Stack AI Engineer
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Denton, TX | Summer internships: full-time June–July 2027 (remote or relocation)
Available to continue part-time during the school year: 10 hours/week, remote

Education

University of North Texas (TAMS) 2025 – Expected 2027
Early college program; Computer Science Track; GPA: 4.0/4.0 *Denton, TX*
- Earned President's List honors in Fall 2025 and Spring 2026; on track for 60+ university credits by 2027.

Open-Source Contributions

oh-my-codex | *TypeScript, AI Agent Tooling* 2026
- Reduced an AI agent prompt by **33%** and fixed stale worktrees through **two merged upstream commits**.

Omarchy Projects | *Linux, Package Manager, VMs for Mac* | PR #195 | PR #331 | PR #769 2026
- Enabled accented typing on Mac keyboards and microphone selection in non-English sessions; repaired tmog-bin update checks across **three merged PRs**.

Products and Projects

AudiLex AI | *Student Software Developer; React, Supabase* | Live Site 2025 – Present
- Shipped a course-grounded voice study platform **used by Northwestern Law students**; recruited through alumni to develop the React interface, Supabase retrieval, and real-time voice experience.
- Cut microphone warm-up wait by **85% (1,000 ms to 150 ms)**; reused capture pipelines, prewarmed voice sessions, and moved citations inline based on **student feedback**.

ScamMap | *Python, Deepgram, Real-Time Audio* | GitHub | Demo Video 2026
- Won **2nd place and $500 at the Finovate Hackathon** for a scam-detection prototype combining live transcription, evidence-backed risk analysis, and repeat-speaker matching.

UNT Grades + VSB | *Core Maintainer; TypeScript, Scheduling* | Grades | VSB 2026 – Present
- Reached approximately **3,400 visitors** and **6,000+ page views** with grade-distribution tools and automated schedule drafting that honors campus, class-time, and lunch constraints.

Computer Vision Research

Computer Vision Researcher 2025 – Present
University of North Texas; supervised by Dr. Heng Fan *Denton, TX*
- Helped build the **1.3M+ frame** VQS-4K benchmark by manually annotating **91K+ video frames**.
- Advancing universal vessel segmentation through **feature extraction, fusion, and multi-feature distillation**.
- Enabled model-assisted annotation by building **TrackGUIv2**, integrating **YOLO/BoTSORT tracking and EfficientSAM** into a LabelMe/Qt interface; managed annotators on a **200K-frame dataset** during a **10-week, full-time paid research appointment** ($4,000).

Research Intern Summer 2025
UT Dallas, Computer Vision and Multimodal Learning Lab *Dallas, TX*
- Added temporal visual context to sound-source separation by replacing the Sound of Pixels image backbone with **3D ResNet** and fusing video features early in the audio network.

Audio-Video Understanding Models | *PyTorch, Transformers* | UCF-101 | LoCoNet 2025 – 2026
- Achieved **88% average accuracy** on AVDIAR active-speaker detection by adapting LoCoNet with VGGish audio embeddings and bidirectional audio-visual cross-attention.
- Achieved **77% UCF-101 accuracy** with a 2-layer, 8-head temporal transformer; froze the pretrained ResNet-50 backbone to limit training to the feature projection and temporal classifier.

Technical Skills

Languages: Python, TypeScript, Java, C++
Computer Vision: PyTorch, OpenCV, YOLO, segmentation, multi-object tracking
AI Agents & LLMs: oh-my-codex, prompt optimization, RAG, Transformers, real-time voice APIs
Full Stack: React, Supabase, Deepgram, Linux + VMs (Mac)